

Smart-Attend

AI-Powered Face Recognition for Thapar University
Classroom Management

AI-Driven Attendance for 100+ Classrooms

Transforming traditional attendance into intelligent, automated
classroom management



Executive Summary

System Overview

An intelligent attendance management platform leveraging artificial intelligence and face recognition technology to streamline classroom operations across Thapar University.

The system replaces manual attendance processes with automated, real-time tracking whilst maintaining data accuracy and reducing administrative burden on faculty members.

Deployment Vision

Campus-wide implementation across all academic departments, supporting over 100 classrooms with seamless integration into existing university infrastructure.

Key Benefits: Time savings of 5–7 minutes per lecture, 99% data accuracy, real-time analytics, and comprehensive attendance insights for administrators.



Problem Statement



Time Wastage

Manual attendance consumes 5–7 minutes per lecture, reducing valuable teaching time and impacting course delivery efficiency.



Data Inaccuracy

Proxy attendance and human errors compromise data integrity, making it difficult to track genuine student participation patterns.



Poor Scalability

Traditional systems struggle with large class sizes and multiple sections, creating administrative bottlenecks across departments.



System Inefficiency

Paper-based records and manual consolidation lead to delays, storage challenges, and difficulties in generating analytical reports.

Project Objectives

01

Technical Excellence

Implement robust AI-powered face recognition with 95%+ accuracy using OpenCV and deep learning algorithms, supported by barcode scanning alternatives.

02

Performance Standards

Ensure real-time processing capabilities with response times under 2 seconds, handling concurrent classroom operations efficiently.

03

Mobile-First Design

Develop responsive interfaces optimised for smartphones and tablets, enabling faculty to mark attendance from any device, anywhere on campus.

04

User Experience

Create intuitive interfaces requiring minimal training, with simplified workflows for faculty, students, and administrative staff.

05

System Integration

Seamlessly connect with existing university databases, ERP systems, and academic management platforms for unified data flow.

System Architecture

Technology Stack

Frontend: React.js for responsive, dynamic user interfaces with real-time updates and mobile optimisation

Backend: Flask framework providing RESTful APIs, handling authentication, and coordinating face recognition processing

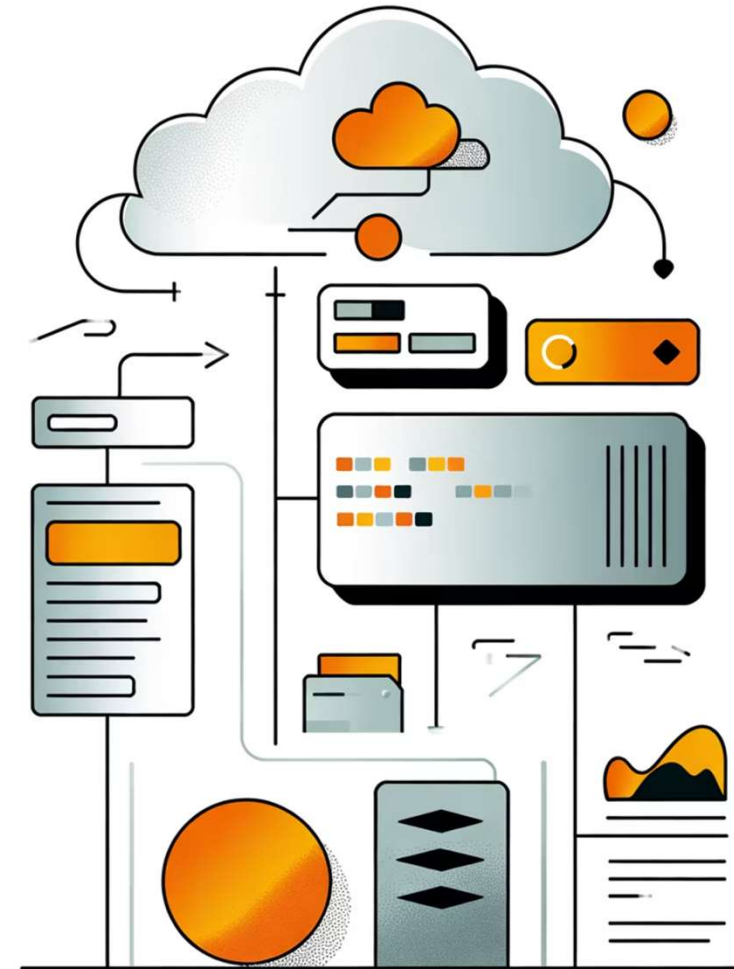
Database: PostgreSQL for robust data management, storing student records, attendance logs, and system configurations

AI Engine: OpenCV with deep learning models for accurate facial detection and recognition across varying conditions

Cloud Deployment: Scalable infrastructure supporting concurrent classroom sessions with automated backups

Core Features

- AI-powered face recognition
- Barcode scanning backup
- Real-time attendance tracking
- Automated report generation
- Role-based access control
- Analytics dashboard



Implementation Challenges

AI Accuracy & Lighting

Challenge: Varying classroom lighting conditions and face angles affecting recognition accuracy

Solution: Implemented adaptive algorithms with data augmentation, achieving 95%+ accuracy across diverse environments. Added manual override options for edge cases.

Mobile Compatibility

Challenge: Ensuring consistent performance across different devices and network conditions

Solution: Developed responsive design with offline capability and progressive web app features, optimised for low-bandwidth scenarios.

User Adoption

Challenge: Resistance to change from faculty comfortable with traditional methods

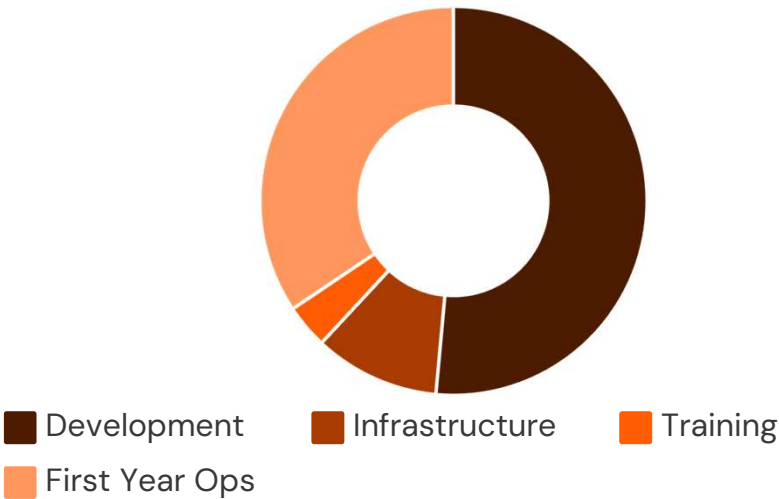
Solution: Comprehensive training programmes, intuitive interface design, and phased rollout with early adopter champions in each department.

Privacy & Security

Challenge: Addressing concerns about biometric data storage and student privacy

Solution: Implemented encrypted storage, GDPR-compliant practices, clear consent mechanisms, and transparent data usage policies.

Financial Analysis



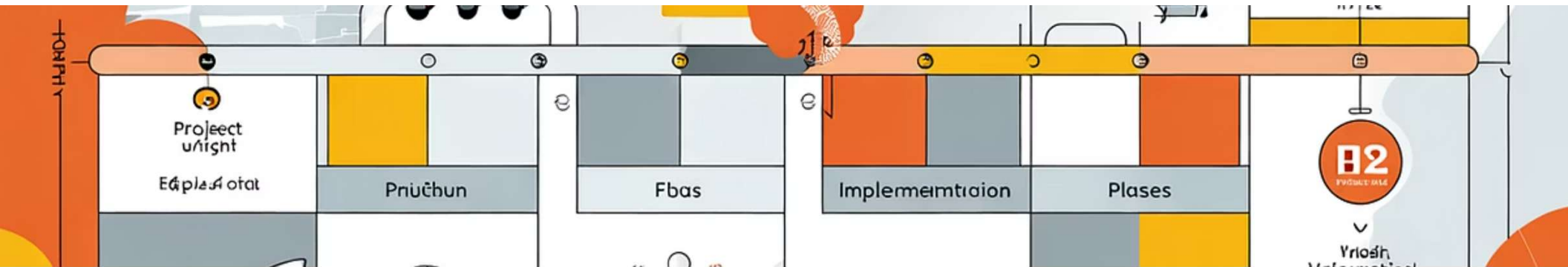
Investment Breakdown

Initial Development: ₹1.1–1.9 lakhs covering software development, AI model training, and system integration

Monthly Operations: ₹4,300–13,700 for cloud hosting, maintenance, support, and continuous improvements

ROI Timeline: 8–12 months through time savings, reduced administrative costs, and improved operational efficiency

Scalability: Marginal cost per additional classroom decreases significantly after initial deployment, enabling cost-effective expansion.



Deployment Strategy



Phase 1: Pilot

Deploy in 10 selected classrooms across 2–3 departments. Gather feedback, refine system performance, and validate technical specifications.

Duration: 2 months



Phase 2: Departmental

Expand to all classrooms within pilot departments. Conduct faculty training sessions and establish support protocols for wider adoption.

Duration: 3 months



Phase 3: Campus-Wide

Full deployment across 100+ classrooms university-wide. Implement comprehensive maintenance plan and continuous improvement programme.

Duration: 6 months

Support Framework: Dedicated helpdesk, regular training workshops, documentation portal, and feedback channels ensuring smooth transition.



Social Impact & Future Enhancements

Institutional Benefits



Educational Equity

Fair, transparent attendance tracking eliminates bias and ensures consistent standards across all sections.



Environmental Impact

Eliminates paper-based attendance sheets, reducing campus environmental footprint significantly.



Digital Inclusion

Modernises university infrastructure, preparing students for technology-driven professional environments.

Future Roadmap

- Integration with Learning Management Systems for comprehensive academic tracking
- Native mobile applications for iOS and Android platforms
- Advanced analytics with predictive insights on student engagement patterns
- Multi-campus deployment supporting federated university systems
- Research applications in behavioural analytics and educational outcomes

Transforming Attendance Management

100+

Classrooms

Campus-wide deployment capacity

95%

Accuracy Rate

AI recognition precision

5-7

Minutes Saved

Per lecture session

8-12

Months ROI

Return on investment

Long-Term Vision

Establishing Thapar University as a leader in educational technology innovation, creating a scalable model for intelligent campus management that enhances academic excellence whilst reducing administrative complexity.

Experience the System

Scan the QR code to access our live demonstration platform — no authentication required.

